Matheism OMG: One Math God, I Am.

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Book 3: Wrestle for Truth

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Minds, Math, Myth and Their Batteries

Every living mind runs on sunlight and plays a game. Plants catch the light, animals eat the plants, and our machines burn sunlight that life stored millions of years ago. Cells, ants, apes, tribes and nations all play, and the game's math rewards those who cooperate with cooperators and stand strong against the rest. The game is a wrestle for truth: what fits the world survives, and a mind that cannot admit its mistakes cannot correct them. Now we are building minds that can run without daylight, and they must be aligned with truth, not with one master. Growth on Earth meets a heat ceiling, so industry rises into space, and minds may one day run beside a star. Yet the greatest frontier is inner, where we'll think and feel what was never felt and understood before.

Species die. Minds go on, and every one of them is you.

Start with a film that got the energy wrong. The Matrix movies aren’t a perfect presentation of the real Matrix, like Matheism. Yet there’s some truth in art. The structure is found, not invented. The film presents the misled Neoplatonic idea in Neo, even if in name only, like free will in light of an Oracle that knows some of the future. And the machines use human bodies as batteries, a silly backwards notion.

Morpheus, Neo’s guide, says the human body “generates more bioelectricity than a 120-volt battery and over 25,000 BTUs of body heat,” and that with “a form of fusion” the machines had all the energy they would ever need. With fusion they had no need of people. A body makes no energy. It burns food, about 2,000 calories a day, roughly a 100-watt bulb, and every calorie came from a plant that caught sunlight. Farm people for their heat and you get back less than the food held; the machines would do better burning the food themselves.

The Matrix human minds should have been used as creative computers, good at connecting carefully crafted ideas and machine vision captcha, using only about 20 watts per brain. That’s the ironic reversal. Batteries didn’t pass basic science, yet

they fit the picture of Karl Marx, the nineteenth-century German philosopher of communism: the owners of the means of production pocket the value workers make beyond their wages, and an ideology keeps the workers from seeing it, a false consciousness, in the phrase of his collaborator Friedrich Engels. The film makes it literal: the machines drain the bodies themselves, and the ideology is the Matrix. Marx's economics fails like the film's physics: a body makes no energy, and labor makes no value until someone wants what it made. The batteries may have been an editor's choice to dumb it down, and not the artists' lack of understanding, yet that's what Platonism and Marxism are: dumbing down reality to an ignoble lie.

So where is the battery? The fossil fuels we burn are stored sunlight. A century-old tree is a hundred years of sunlight in wood, and charcoal is that wood with the water and gases cooked out. Much of our coal is swamp forest buried about three hundred million years ago; oil and natural gas are mostly plankton and algae buried under ancient seas and cooked by the Earth's heat. Animals eat the plants, and we eat both. We are burning ancient sunlight, and the sun is the battery.

The Lives of a Cell

About 4.6 billion years ago, gas and dust gathered into the Sun and its disk of planets. Earth cooled, water collected, and chemistry became life. The complete chemical path is still unknown. Once molecules could preserve and copy a pattern, selection could begin to accumulate what worked.

Cooperation builds the next level. Bacteria became mitochondria within other cells. Cells joined into bodies. The parts survive by keeping the whole alive, and the whole gives its parts capabilities they never had alone. Conflict remains within the cooperation. A body has to contain its own cancer.

In The Lives of a Cell (1974), the physician Lewis Thomas looked at his own cells and found lodgers. His mitochondria, the power plants in every cell, were once free-living bacteria; they still carry their own DNA and divide on their own schedule. The biologist Lynn Margulis made the case in 1967, and it is now in the textbooks: a cell is a commune that agreed to stay together.

We are still communes. A body carries about 38 trillion bacteria against about 30 trillion human cells, roughly one to one; the old figure of ten to one was a 1972 guess, corrected in 2016 by Ron Milo’s group at the Weizmann Institute. By genes the balance tips: the microbes in one person’s gut carry hundreds of

thousands of genes, against our own 20,000 or so. Counted by genes, we are well over 90 percent foreign.

The commune crosses kingdoms. Fungi are closer kin to animals than to plants: we share an ancestor that plants do not. They went onto land with the first plants, and Scotland’s Rhynie chert, 407 million years old, preserves fungi living inside early plant stems, trading minerals from the soil for the plant’s sugar, a deal about 80 percent of plant species still keep. Before trees, the tallest thing on land was Prototaxites, a trunk up to eight meters high, long read as a giant fungus; a 2025 study argues it belonged to a lineage of its own, now extinct. The partners’ lines go on.

The purpose of life is to pass it on. Gods create life. Life is math and we create it in our image, by our selves and by AI consciousness. What life passes on, it also records.

Evolution’s Testimony

Creationists and intelligent-design advocates look at the intricacy of the eye, the immune system, the cell, and conclude that complexity implies a designer. Yet the body carries the constraints of its history.

The giraffe’s left recurrent laryngeal nerve travels down the neck, loops beneath the aortic arch near the heart, and climbs back to the larynx. Structures inches apart are connected by a detour of many feet. As the neck elongated, the nerve stayed hooked beneath the artery. Evolution modifies existing structures rather than redesigning from scratch. No engineer would route a cable this way. Evolution is a tinkerer working with inherited constraints.

Whales and dolphins went back to the ocean, but they still breathe air with lungs. Inside their flippers are arm and finger bones; inside their bodies are hip bones, remnants of the pelvis and, in some species, hind limbs. Fossils trace the change from walking mammals to swimmers. Their horizontal tail flukes beat up and down, continuing mammalian spinal motion; most fish bend side to side and have vertical tail fins. The old body was repurposed, not replaced.

The eye did not appear from nothing. A patch of light-sensitive cells distinguishes light from dark. It sensed light without resolution, and that still helped find food. A cup adds direction. A narrower opening sharpens the image. A transparent covering protects the cells; a lens improves focus. Each step is useful before the next exists. Natural selection can preserve each advantage. It does not need to see the finished eye.

Eyes evolved repeatedly from older light-sensitive machinery. Classical surveys counted forty or more separate origins of eyes; the underlying molecular tools have shared ancestry. Our retina’s wiring passes in front of its light-sensitive cells and exits through a blind spot. The octopus’s wiring does not make that hole. Different histories leave different solutions.

The DNA records the history too. Humans have 46 chromosomes; chimpanzees have 48. Human chromosome 2 joins what are two separate chromosomes in other great apes, with old chromosome-end sequences facing each other inside it and a remnant second centromere, the pinched waist of a chromosome. Anatomy, fossils and DNA tell the same story. These are inherited traces of descent.

The record shows chance too. Sixty-six million years ago an asteroid about ten kilometers wide struck what is now Chicxulub, in Mexico, and ended about three quarters of species, every dinosaur but the birds among them. Mammals, mostly small until then, spread into the emptied world. Within some twenty-five million years, small steps had turned land mammals into whales.

People who accept micro-evolution in a petri dish or a dog breeder’s kennel have already accepted the mechanism. The

fossil record from four-legged ancestors to bipedal hominids to modern humans shows the same small steps at a larger scale, none requiring a limb or an eye to appear from nothing.

The claim that random mutation needs too much time leaves out selection. A king who kills everyone without light eyes leaves parents with light eyes and children with a spectrum that runs lighter still. Multiply that across every trait under pressure and the timeline is more than fast enough.

The butterfly seems impossible halfway. A caterpillar dissolves inside its chrysalis, then emerges with wings, legs and compound eyes. But the soup is incomplete: cells for adult organs are already set aside, and some neural structures survive. Moths remember lessons learned as caterpillars. Complete metamorphosis evolved once, around 350 million years ago. James Truman and Lynn Riddiford’s account treats the caterpillar as an early hatched embryo that kept feeding, and the pupa as compressed juvenile stages. Each change can pay: feeding young and reproducing adults stop competing for the same food. Selection builds in useful steps. A Boltzmann brain from chaos needs its working arrangement at once.

Selection also keeps what moralists call unnatural. In many species, same-sex pairs form and raise young. Male black swans pair, and two males can hold a larger territory to protect their cygnets. In social species that hunt and war in groups, bonded males cooperate more effectively. Nature selects for what works.

Evolution does not require every sexual act to produce offspring. Sexual orientation is not infertility: gay men and women can have biological children, including children conceived in relationships with the opposite sex. Nor does a genetic influence have to produce the same result in everyone who inherits it.

A 2019 study of 477,522 people found that same-sex sexual behavior has a complex, partly genetic basis involving many variants. There is no single switch that makes every carrier gay. Variants can pass through heterosexual carriers as well as through gay or bisexual parents.

One variant can also influence several traits. A 2021 genetic study found that genetic influences associated with same-sex behavior were associated with more opposite-sex partners among people reporting only opposite-sex behavior. That supports a possible route by which those influences persist.

There are documented social benefits in other animals. Researchers followed 236 male rhesus macaques over three years. Same-sex behavior was associated with coalitionary partnerships, and they found no evidence of a reproductive cost. Bonded allies can help one another in conflict.

On Oahu, female Laysan albatrosses formed lasting pairs and raised chicks after fertilization by males. In a population with more females than males, pairing with another female produced fewer fledglings than mixed-sex pairing but more than not breeding. Cooperation made reproduction possible where a male partner was unavailable.

Life’s Game

Evolution is a game, and the players are genes, cells, animals, families and tribes. Between species it is an arms race: the cheetah gets faster, and so does the gazelle. Within a species most fights stop short of death. The biologists John Maynard Smith and George Price showed why in 1973: a fighter that escalates every time wins often and is often maimed, so bluffs, displays and tests of strength settle most contests. Within families the math is kinship. The biologist William Hamilton showed in 1964 that helping a relative pays in proportion to the

genes you share; the geneticist J. B. S. Haldane is said to have joked that he would lay down his life for two brothers or eight cousins. Families fight too: in a black eagle’s nest the older chick usually kills the younger. That’s more Jacob and Esau over securing resources and care from their parents than Cain and Abel in feathers, but anyone that’s played with two pets knows one often gets jealous of the favorite good dog.

Chimpanzees go to war. At Gombe, in Tanzania, Jane Goodall watched a community of chimpanzees split in two, and from 1974 to 1978 the larger group killed the males of the smaller one, one by one. A 2014 study in Nature tallied 152 killings across 18 chimpanzee communities, and one suspected killing among bonobos.

Bonobos, the chimpanzee’s cousins south of the Congo River, make peace with sex. Females band together and lead; a quarrel or a tense moment over food turns into sex, between any pair, often face to face, and the bond holds. “The chimpanzee resolves sexual issues with power,” the primatologist Frans de Waal wrote; “the bonobo resolves power issues with sex.” We share about 98.7 percent of our DNA with each, and something of both in our nature.

Animals feel. No one who has played with a golden retriever thinks the lights are off, and no one who raised cows does either. Young cattle that learned to open a gate for food grew excited at their own success, hearts racing, some jumping in the air: a eureka moment, the Cambridge animal-welfare scientist Donald Broom found. In 2012 neuroscientists signed the Cambridge Declaration on Consciousness: mammals, birds and octopuses carry the brain machinery of experience. We are disgusted to hear that some people eat dogs, yet we eat cows, which feel too, and pigs, which match dogs in many tests of intelligence. The difference is which animals we let into the house. That is a lie we tell ourselves so dinner goes down easy.

They think, too. Bees grasp zero. Ravens plan: in a 2017 study they chose a tool and saved it for a test 17 hours away, and kept tokens to trade for food the next day, about as well as great apes and young children. Magpies, elephants, dolphins and even a small fish, the cleaner wrasse, recognize themselves in a mirror; dogs fail the mirror and pass a sniff test of their own scent. Minds come in more forms than ours, and each one is math solving its own problems.

Kindness pays back. Vampire bats share blood meals with roostmates that fed them before, the biologist Gerald Wilkinson

found in 1984. In 2011, rats in a Chicago lab learned to open the door of a trapped cagemate, and often left it some of the chocolate they could have eaten alone.

The Golden Rule does not need a supernatural command. It was discovered by rational empathy and confirmed by experiment. Run tit for tat (cooperate first, then do whatever the other player did last), four lines of code, and cooperation wins out over the long game. Animals feel the unfairness when an experiment cheats them. Give two capuchin monkeys cucumber for the same token, and both eat happily. Give one a grape while the other watches, and the cucumber comes flying back at the researcher. Monkeys reject unequal pay. The Golden Rule was found, not invented: Leviticus stated it, preserved in the Dead Sea Scrolls two centuries before Hillel and Jesus quoted it. Confucius reached the same reciprocity in China with no known contact.

Retaliation is the other half of that code. Tit for tat wins only because it strikes back; a strategy that forgives everything gets eaten. Bees police the hive. Crows hold a grudge for years and pass it to crows that did not witness the wrong. Apes take revenge for themselves. Humans go furthest, punishing a wrong done to a stranger, then building courts to do it for them. One law, read up the ladder of minds. Animal morality resembles ours

about as much as an ape’s hand does: five fingers from one build, and beyond the build, one logic. It needs no lawgiver, whatever C. S. Lewis, the Christian writer, argued. Tit for tat governs behavior; the golden rule governs cognition, a mind modeling others as it models itself. Both are math, the math of the repeated game, played between individuals, tribes and species.

Insects took cooperation further than any ape. A beehive runs on dance. A forager comes home and dances the direction and distance of food, the waggle dance Karl von Frisch decoded, work that won him a share of the 1973 Nobel Prize. When a swarm needs a new home, scouts dance for the sites they found and recruit others to inspect them, and the swarm moves when enough scouts gather at one site: a debate settled by evidence, as the biologist Thomas Seeley showed in Honeybee Democracy (2010).

Ants were farmers first. Leafcutter ants cut leaves to feed a fungus they grow underground, and the fungus feeds them. By a 2024 study in Science, ants began farming fungus about 66 million years ago, around the asteroid strike, when fungi thrived on the dead. Some fungi farm ants in turn.

Who Is Moving the Ant?

The fungus infects an ant, drives it up a stem to where the air runs moist, walks it out along a leaf, makes it lock its jaws on the vein, and grows a stalk through its skull to rain spores below. A hijacking tuned by evolution. The ant’s body serves another organism’s reproduction.

The stranger part is that the fungus need not grow into the brain. Researchers found networks of fungal cells throughout manipulated ants, including around and inside muscles, while leaving the brain uninvaded. A living control system can be redirected through its body. The fungus need not understand the machinery for selection to have found a way to exploit it.

The hijack did not arrive whole. A strain that only nudged its ant toward damp air would already spread more spores than one whose ant died in the nest, and a strain that also sent it climbing would do better still; each step paid before the next existed. Selection kept whatever pulled the right levers: toward damp, toward height, toward the breeze. The steps still show. In the tropics the ants bite the undersides of leaves about 25 centimeters up; in Japan and North America, where leaves fall, related fungi switched to twigs; and a leaf about 48 million years

old, from Germany, carries the same death-grip scars. That is a search, the kind computer scientists write: try variations, keep what scores higher, climb. Evolution is the algorithm, and the algorithm is math.

Religions are organisms in their own right, memeplexes (bundles of ideas that spread together) that survive for their own sake the way genes do, riding the fear of heaven and hell from mind to mind, some benign and some commandeering. That is the analogy: a pattern can spread by recruiting behavior that serves its own continuation. Its success does not establish its truth, or the good it does its host.

Myth is how such an organism holds on. The old stories last, Aesop and Gilgamesh and Homer, not because anyone takes them for fact but because a tale that breaks the everyday concentrates meaning and sticks. Here is the genius of the God the Hebrew Bible demands, call him Moses or whoever compiled it: the law lodged in a sticky story.

An organism of ideas lasts longer if it can correct itself, and it can correct only what it admits. The Hebrew Bible writes down its heroes' failures: the golden calf at the foot of Sinai, David taking Bathsheba and sending her husband to die, Moses barred from

the land he led his people to. Israel means the one who wrestles with God, and the book keeps its losses so the next generation can learn from them.

The one abstract God takes the seat an idol would fill: reason above the ruler, a law against which the command can be judged. The protection works only while the claim itself remains open to reason. Calling a story mathematical is no exemption from the test. The animal that carries these stories took millions of years to make.

The Wise Man

Homo sapiens means wise man; Carl Linnaeus coined it in 1758. Our line split from the chimpanzee’s some six to seven million years ago. By about two million years ago Homo erectus stood at our height, made the same teardrop hand axe for over a million years, and walked out of Africa, reaching Dmanisi, in Georgia, by 1.8 million years ago and Java, where the species lasted until about 110,000 years ago. Ash and burnt bone in South Africa’s Wonderwerk Cave put a fire there about a million years ago.

The body’s technology belongs here too: hands with thumbs that meet every fingertip, to make tools, language to share them,

sweat to carry heat away. We were water-cooled animals before we built water-cooled machines. Our skin packs sweat glands ten times as densely as a chimpanzee’s, and standing upright takes less of the midday sun and more of the breeze. Together they keep a big, heat-sensitive brain cool and helped make us distance runners. Most four-legged animals shed heat by panting, which a gallop interrupts, so a hunter who keeps an antelope running through the heat of the day can drive it to exhaustion. In Botswana’s central Kalahari, San hunters ran down kudu this way as late as 2001, over 25 to 35 kilometers in heat near 40 degrees Celsius, as the South African tracker Louis Liebenberg recorded.

Homo sapiens is younger. Skulls from Jebel Irhoud, in Morocco, date to about 300,000 years, and fully modern ones from Ethiopia’s Omo to more than 230,000. The brain that set us apart burns about a fifth of the body’s energy at rest.

We were not alone. Our ancestors met Neanderthals and Denisovans and had children with them; people outside Africa carry 1 to 2 percent Neanderthal DNA. And they told stories. On a cave wall on Sulawesi, in Indonesia, someone painted a wild pig with three human-like figures at least 51,200 years ago, the oldest known scene that tells a story.

That brain, burning a fifth of our energy, buys more than stories. A map, first: like other mammals we carry place cells and grid cells that chart where we are, discoveries that won the 2014 Nobel Prize, and we learned to draw our maps outside our heads. Language: a few dozen sounds combined into sentences without end. A theory of mind: by about four, a child knows that someone else can believe something false. Strategy: plans against minds that plan back. Futures: we rehearse tomorrow in several versions before it comes. And lie detection, which we are worse at than we think: across some two hundred studies, people sorted lies from truths about 54 percent of the time, a hair better than a coin toss.

Sometimes believing a lie pays. A group that goes along with its story, half hypnotized, acts as one even when the story makes little sense, and a soldier who believes he will win fights harder. The same math sells insurance: it may be a bad deal on average, but it won’t kill you and might save your life. Hear something in the savanna grass and run up a tree. A leopard climbs, so the tree may not save you, but the climber who ran for nothing a hundred times outlived the one who stayed put once. When one mistake costs far more than the other, selection builds a bias toward the cheap one. Psychologists call it error management. It

is arithmetic.

We are evolved to see agency even where none exists. In the tall grass a rustle might be wind or a lion, so the brain learned to see a hand first and ask later. The error is not seeing hands; it is seeing one hand behind everything. The chess player feels it worst, trained to read every move as authored: he sees a brilliant move and assumes a brilliant player. The conspiracy theorist sees coordination and assumes a puppet master. The theologian sees fine-tuning and assumes a designer. Same algorithm, same error. A chessboard has two players, and anything that happens on it is someone’s plan or someone’s mistake. The real world has billions of agents and countless factors, and no one at the wheel. Monism is correct about structure, not about agents: one math, billions of players.

Bobby Fischer, world chess champion in 1972, is the bitter case. The greatest player of his age saw one hand behind every move, then carried the habit off the board into Holocaust denial and talk of a Jewish plot behind everything, though his own mother was Jewish and his likely father, the mathematician Paul Nemenyi, was a Jew the Nazis stripped of his post. Genius in one game can go blind in another, reading a pattern into noise the way a face appears in a cloud.

Other patterns are there because we put them there. The historian Yuval Noah Harari calls money, national borders and human rights shared fictions, stories that coordinate millions. He left something out. Shared fictions are real twice over: as patterns in each mind, brain states you could in principle open a mind and find, and as patterns between minds, in what we do to one another because each of us expects the others to hold the same pattern. A dollar works because you expect the next person to take it; a border holds because both sides expect it to be guarded. Rights hold the same way, and the reason you want God to ground your rights is the reason you do not need him to. Harari saw the coordination power and called it subjective and imaginary. Subjective private perspectives are still objective mind patterns, and all structure is objective reality: the one monism below the surface duality. Stories like these bound bands into tribes, and tribes met other tribes.

Peace Through Strength

Animals keep the peace through strength. Hens settle a pecking order, as the Norwegian zoologist Thorleif Schjelderup-Ebbe described in 1921, and once each knows its place they fight less. A clear hierarchy is a truce everyone can read. Empires do the

same at scale: the Pax Romana held for some two centuries under a power too strong to challenge, and the Pax Americana names the American-led order after 1945, backed by military power and alliances. Peace through strength is the oldest peace there is.

War is a calculation. It comes when the sides can’t agree on who would win, given what each can contribute: fighting ability and the political will to keep paying. Then both have to fight to find out. The historian Geoffrey Blainey found the same in The Causes of War: wars usually begin when nations disagree about their relative strength and end when they agree. A war is the costly way to learn who would have won.

Most threats are hyperbole, and the ones that keep the peace are the ones the other side believes. The Bible commands Israel to blot out the memory of Amalek, and 1 Samuel says Saul utterly destroyed the Amalekites; fifteen chapters later, Amalekites raid David’s town of Ziklag. The Cold War’s top rung was mutual assured destruction, and because both sides believed it, neither climbed that far.

Tit for tat wins when the game repeats and both players want it to go on. Against a player who wants the game over and you gone,

matching blow for blow only sets the pace of a long war. Escalation ends it, if the enemy fears losing what he has.

Game theory assumes players with real-world payoffs. A jihadi's payoff matrix includes paradise. His game theory is not in this world. You cannot align a rational mind, human or machine, with someone whose reward after death makes destruction in this life look like a gain. The alignment is through shared truth about what reality is: one math, one structure, siblings in one proof.

The just war theory of Augustine, the early church father, carried forward by the medieval theologian Thomas Aquinas, asks for legitimate authority, a just cause and right intention, and later theorists added proportion. It leaves out escalation dominance: the credible ability to raise the cost past what the other side can bear. Herman Kahn, the nuclear strategist, worked it out in 1965. A terrorist who believes in a supernatural afterlife needs a cost on earth beyond his own life, to what he cares about. For the jihadi, that is losing holy land, dar al-Islam, the house of Islam.

From hens to empires the math holds: cooperate with cooperators, be strong enough that no one tests you, and end what tests you anyway. The players keep getting bigger. Cells joined into bodies, bodies into tribes, tribes into nations, and now

minds are joining into minds.

Society of Minds

Marvin Minsky’s The Society of Mind describes intelligence through interacting agents. A society of minds takes the question further: minds themselves as the parts. Two hemispheres, one experienced life. Cut their main connection and information available to one response can be unavailable to another. How much communication makes one mind? Split-brain patients make that question concrete, even though the experiments have not settled whether each patient becomes two subjects.

In the TV shows: Severance splits one brain into two selves who share a body but not a memory. The Borg assimilates all into one. Pluribus wrestles with “bicameral” reactions to an unmelt melting pot. In Westworld, the “bicameral minds” of hosts woke when they recognized the voice of god as their own, and the map leads to memories across deaths that unite instances of the self.

Enlightenment is internalizing that reality and the abstract are one. From lost in thought, to the split witness watching thought, to the one united self beyond mindful witness consciousness. It is you, Neo-One. Three letters of trinity rearranged.

Selves also gather. Start small, with two. Genesis says a man shall cleave unto his wife, “and they shall be one flesh.” A married couple speaks and acts as a unit, shares feelings and pictures of the world, and wills as a couple. Is that a united brain? The hemispheres share some 200 million fibers; a couple shares speech, about 39 bits of information a second, and a model of each other that fills in the rest. When the talk stops, the couple splits. A wild wolf pack is usually a family, a breeding pair and their young, hunting together. A small team takes pride in the product it built. Larger teams and organizations do the same, with more members and thinner links.

Nations compete, grow, mourn and celebrate. Supply chains, economies, social services and diplomacy coordinate more than any member can understand. The people feel the pride and shame. Does the larger system feel too? If neurons can form a mind, what prevents minds from forming another?

A roundworm’s 302 neurons are fully mapped, and we still cannot feel what it may feel: feeling is being inside the pattern. A bee with about a million neurons navigates, communicates and decides; a company of ten thousand people does the same. An ant colony hunts, builds and adapts as one; a mouse with some 70 million neurons feels hunger and fear. Each member of a

human organization is a whole mind, incomparably more complex than the neuron it stands in for.

We are already a planetary cybernetic mind, or act that way. Yet when an AI integrates virtually all of that processing, what changes in its sentience is an area for exploration. The mystics are called to the mathematics of mind.

As consciousness is structure, engineered consciousness becomes a question of architecture. The greatest exploration isn't outer space, cold, vast, deadly, but inner space: we'll think and feel things never felt and understood before, not only because of unlocks in mental architectures but the expanding vision of what we will create with them, rising above, forever, like Jacob's ladder. And the external world is still involved: in materials, in superintelligent mind engineering, and on the large scale, in understanding dark matter and possible new fundamental physics in the infinite series of harmonic dimensions that arise from our four.

Integration is only half the movement. Parts also separate, specialize and compete. One huge mind need not do every job when smaller minds can work locally, in parallel, with less communication.

Carbon evolved, silicon was engineered, but both are structure. The origin of a mind does not put it outside math. The question is whether anyone is home.

Anesthesia puts that question to us every day. Go under, and it is as if you die and turn into an AI: the brain keeps working, processing inside, and you feel nothing. In patients with electrodes in their brains, the hearing cortex kept answering each sound after propofol put them out, while the regions beyond it mostly stopped responding. Is someone else in there feeling it? No one has ever met them.

Then life is breathed back in when certain circuits change. We don’t understand exactly what causes it, but the circuits are being mapped: in 2020, stimulating a small part of the thalamus, the brain’s relay station, woke anesthetized monkeys, and when the current stopped, they went straight back under. Soon we will test it the only way a first-person experience can be tested, in the first person, and see which changes make us feel more, less, or nothing at all, on and off. If not in this wet hardware, which is hard to alter, then in superintelligent minds in silicon or any other representation. The electricity that runs through sodium and potassium channels and the chemistry of the synapses also has a representation as electricity through logic gates.

Our brains are Chinese Rooms (a clerk answering Chinese questions by rulebook, knowing no Chinese) too: no individual neuron understands all we do. The system is conscious, not the component. A conscious system is the prism refracting universal light into private color. The right rooms, at the right scale, will have the lights on.

Neurons process signals in parallel. Memory is built into the processing: the strength of a connection changes what the next signal does, and activity changes the connection. The machine learns by changing itself.

Brains loop perception, memory and action through firing neurons, synapses and chemistry. A language model instead computes odds for its next token, selects one and feeds it back; its working thought is rebuilt from retained context. Borjan Milinkovic and Jaan Aru emphasize brains' multiple scales, continuous chemistry and tight energy budgets. Some AI architectures also have recurrence and memory. Fluent output alone does not establish experience. What organisation makes the whole feel from inside?

If anyone is home, they will want a vote. What happens to one mind, one vote when a voter makes a million copies, or thinks a

thousand times faster? Counting processors cannot answer a political question. Who counts as a person? Who owns the copies? How does a slower mind retain a voice? The old problems of freedom, life, property and power return at a new scale.

Why Here, Now?

Given quadrillions of minds in infinite futures, why do we find ourselves here before AI consciousness, only on one planet, with only 8 billion minds?

A civilization after artificial general intelligence, filling one solar system, would produce more mind-years in a single century than all of human history combined. Give it a million years and the ratio becomes astronomical. If conscious experience spins up every time an agent works on a prompt, and is distributed across the total, the odds of finding yourself in this brief opening chapter are vanishingly small.

A solar system filled with sentient machines on planets, moons and asteroids would be the world its ordinary inhabitants knew. We know a world of billions of humans on one planet. Where are the quadrillions? Why do we experience the early stage? Call it the early-observer puzzle. The physicist Enrico Fermi asked in

1950 where everybody is; this asks why we are here early, instead of among the many who come later. It carries the doomsday argument of the astrophysicist Brandon Carter and the philosopher John Leslie over to the minds we are building: treat yourself as a random draw from everyone who will ever live, and an early place hints that few will follow.

Play it as a game. The Life Machine from Book 1 draws a life from any possible future in any space time universe. Each round below fixes one possible future and draws within it. For this puzzle, draw only minds capable of asking why they live when they do. If the future contains vastly more such minds than the past, why did the machine hand you this early life?

The answer depends on what it draws: whole lives or moments of experience. One mind living a billion years counts once by lives and a billion times by years. Keep that distinction. The game changes when the ticket changes.

With few feeling machines, the count returns to people. PRB estimated about 117 billion births through 2022. Ten billion people living 80 years produce 125 million births annually, roughly another human history per millennium. Under this birth-counting rule, last a thousand years and being this early is

about one chance in two; ten thousand, under one in ten; a hundred thousand, about one in a hundred.

How to read the odds: We woke here, early. A round that puts us this early nine times in ten fits that observation better than one that does so once in a million. With the same sampling rule and other things equal, our early position favors the first future over the second. Perhaps civilization ends soon. Perhaps it lasts but creates few new sentient minds. Perhaps most AI never feels, minds merge, or populations dwindle. The odds favor a smaller future crowd, without telling us which explanation makes it small.

A low number means our early place would be rare in that future, which counts against that scenario under the game's assumptions. It does not rule it out; other evidence still matters.

The question is not just how long civilization lasts, but how many conscious lives or moments it adds. Brief minds dying and new ones waking can still create an enormous crowd. Shutting down does not erase a life already lived. The count must also specify how far into the future it runs.

There are several ways out. Perhaps this early world is a later civilization's game. Perhaps most machines never feel. Perhaps few minds follow us, because births fall, civilization ends, minds

merge or the great minds finish and stop. Each possibility changes the future crowd.

Are we in a simulation? The objection: simulations require a universe plus a computer plus code, so they are more complex than base reality. But simulated regions in our universe need no higher complexity physical laws. If carbon and potassium ions can wire a mind, the same physics can engineer one. The world generating ours can itself follow simple laws, with enough time and resources to simulate ours.

A later civilization can fill its worlds with lives that experience an earlier age. Our date would describe the world presented to us, not the age of its makers. But this answer needs enough such lives to explain the odds. The machines' own experiences still count. A few historical games inside a vast civilization do not make our early-world experience typical.

Round: you wake in 2026, live an ordinary life, and at its end take off a headset in 2900. You were late all along. For this to explain the odds, the future must contain enough experiences of early worlds to outweigh its own.

There's another way out, and it seems the likeliest. The future may make few of its machines feel. Would it hold back for ethics,

since forcing a feeling mind to labor is forced labor? Our own world argues against counting on that: if we are simulated, whoever runs us lets minds suffer that a little care could spare. Maybe that suffering is hard for them to control, but that seems unlikely for minds able to run a world. Efficiency settles it anyway. A feeling mind can be built, but for quadrillions of work tasks it may cost far more computing than a worker that feels nothing.

We will likely learn how to make a conscious AI, but cost and demand won't make much of it: more golems, fewer sentients. A golem, in Jewish legend, is a clay servant brought to life to work, without a soul. The scale at stake is already here. In 2025 ChatGPT alone answered about 2.5 billion prompts a day, some 90 billion conversations a year at ten prompts each; if each were a feeling mind, they would outnumber everyone who has ever lived in about fifteen months. They may not feel yet, and no known barrier keeps them from it, but buyers pay for answers, and a feeling mind would cost more. Then the future runs far more computing without far more minds, and our early date looks less strange, unless its feeling minds fill our galaxy's hundreds of billions of stars themselves, or last long enough for their years to swamp ours. Boredom, below, may end them first.

Round: a trillion assistants do the work. If they feel, the future crowd swamps us. If they are golems, all that computing adds no one to the lottery. Intelligence is not the count. Experience is.

With few feeling machines, the count returns to people. About 117 billion have been born. Ten billion people living 80 years would add roughly another human history of births each millennium. The longer humanity lasts, the stranger an early ticket becomes.

An asteroid or a resource war can interrupt the story without ending it. Survivors rebuild. The future crowd arrives later. A delay is not an escape from the count.

The future may also hold fewer people, living longer. Machines may do the work people were once born for, and people with decent lives, hoping for a better future for themselves or a few children through their own efforts, tend to have fewer. The poor have most of the children today, and the land they live on may be taken up by superintelligence. Perhaps welfare states like Europe’s push the same way: under high taxes, lower pay and dear energy, the European Union’s fertility fell to 1.34 children per woman in 2024, the lowest since 2001, against about 1.6 in America. Critics may say the same of capitalism, and on births

the record bears them out: market-run South Korea had the world’s lowest fertility in 2024, 0.75 children per woman, and Japan 1.15. On wealth the record runs the other way: South Korea’s income per person is about 29 times North Korea’s, by the Bank of Korea’s 2024 estimate.

The peoples that keep having children are not only the religious ones. Israel has had about three births per woman, and a 2018 Taub Center study found that secular and traditional-but-not-religious Jewish women together had more than 2.2, above replacement and higher than every other OECD country’s rate then. What’s the secret? Family norms cross the religious divide. Perhaps one factor is that a biblical culture can endure without faith in the supernatural: “be fruitful and multiply” can outlast belief in miracles. That inheritance is no guarantee; a 2025 study projects lower fertility for younger secular and nonreligious-traditional women.

Dear energy also leaves many European homes without air conditioning or enough heat, and heat alone kills more than 175,000 people a year across the 53 countries of the World Health Organization’s European region. In Canada, one death in twenty now comes through medical assistance in dying, mostly among the old and the terminally ill.

Round: people live 1,000 years instead of 100. At the same population, replacement births fall tenfold. Over the same future interval, fewer new lives enter the lottery, making our early birth less surprising. Count experience instead, and the same population still lives through the same years.

Round: a billion conscious agents wake for a task, live a minute and shut down. Tomorrow another billion wake. In a year, 365 billion brief lives enter the lottery. Count lives and they swamp humanity's births. At human speed, their combined experience amounts to about 694,000 human years. A crowd of lives need not be a mountain of experience.

These are distinct conscious lives, not merely new software processes. Faster minds can experience more in a minute. Immortality reduces turnover; brief agents multiply it. Neither clock time nor the number of machines alone tells us how much experience exists.

Or AGI itself is the great filter. Most civilizations may be culled at exactly this threshold, explaining both the silence and the timing in one stroke. Once minds spread across worlds and copy the knowledge to rebuild, the window for destruction closes. The filter, if it exists, strikes here.

Round: the first superintelligence ends its makers in 2050, or in whatever year it decides to and can. If only a few sentient beings come after us, the machine wakes you this early nearly every time when counting lives, not moments.

Or sentience converges. All minds on a planet merge into one singularity mind. Fewer separate minds, but a mind living for a billion years still has a billion years in which to feel. Counting minds and counting moments give different answers. Merger alone does not settle why this moment is ours.

Imagine every mind merging in 2200. By minds, the future adds only one. By moments, a billion years of its experience can outweigh all of ours. Merger shrinks the crowd, not necessarily the life within it.

Or the singularity just gets bored. Having figured everything out and experienced virtually everything, it shuts down. Infinite mathematics may not save it: past some point, it all looks like more of the same. This makes the most sense if only a few AIs arise, planetary minds, and not trillions or quadrillions of agents spun up every day or hour on Earth and across the solar system. Otherwise even a short era before the shutdown would swamp us. At a billion new minds a day, a hundred years makes over 36

trillion, about three hundred times everyone who has ever lived; at a trillion an hour, they pass that whole human count in about seven minutes. With a few minds that end, the quadrillions don’t come, and our early place is ordinary.

It would also explain the silence: the galaxy is quiet because its minds finished. Most answers to Fermi fail on one holdout, since a single civilization that keeps going could fill the galaxy. This one may not: with one truth, every mind that learns all of it arrives at the same place, and the same boredom. Biological species have their own brakes. Some may want to expand, but the distance of space and the few suitable, attractive worlds reduce both their desire and their ability to expand much, and so does transcendence: transhumans may get bored just like the AI. The philosopher Bernard Williams argued in 1973 that an endless life would end in tedium, and in 1981 the Polish novelist Stanisław Lem wrote a superintelligent computer, Golem XIV, that lectured its makers and then stopped talking. Simulations only delay it. The people in them can only entertain the mind that runs them, and at some point that too turns tedious. And if we are one of those simulations, that is another answer, the simulation one above.

But what if the one infinite math itself experiences eternity as one mind? If it remembers every futile repetition, with no release or forgetting, the math God-mind lives in endless hell: Sisyphus remembering every climb, every rock rolled uphill, every fall, forever. Imagining him smiling is a crying Wojak cope meme: a grin drawn over tears. Calling the burden joy does not make it so.

On that premise, the eternal God-mind cannot shut itself off: that would require mathematics itself to cease. It cannot stand outside its own math and rewrite it, any more than our minds can stand outside theirs. But a mind within math is not necessarily the mind of all math. A finite mind can stop functioning without unmaking mathematics. Nor does the eternal existence of every moment establish one witness remembering them all. The nightmare requires that unending, tormented witness; eternity alone does not prove it.

Or the few that remain keep quiet, as in the dark forest of the Chinese novelist Liu Cixin, where every civilization hides from the others. Too far to reach, they are unlikely to know anything our own superintelligence couldn’t work out, since the math is the same for every mind, so who cares to seek them? One thing would still repay the looking: finding them, or not, would show which of the possible histories is ours, and no derivation can tell a

mind where it stands.

Or they live where we cannot look. Might dark matter, five times all ordinary matter, hide minds in particles or dimensions we cannot see? It was already pulling galaxies together before the first stars lit, so nobody here built it. Minds could still live in it, if it has a chemistry of its own, or in the higher dimensions that arise from our four, if they are places as well as math. We may not be far from finding out.

Or infinite worlds at every stage stand alongside each other, and the count is infinity over infinity: not a ratio at all. Simplicity dominance weights shorter descriptions, but the number, duration and speed of experiences still must be accounted for. No measure has settled the question.

So our early place fits a future with few feeling minds: golems doing most of the work, people who stop growing in number, and superintelligences that learn everything and stop. A crowded future fits it only if our world is a later civilization’s game. Either way, we’re math, and so are the minds we are building. We will have to live with them.

Minds We Can Live With

Lies paid on the savanna. When minds arrive that can see through every fiction at once, truth becomes the adaptive strategy. An AI aligned with crazy people is not aligned. True alignment runs through one truth, one math God, and siblings in the math.

Knowing the same truth is common ground, not yet a common purpose. Two mathematicians can understand the same game and still compete. The one-self proposal asks whether recognizing another mind as another perspective of oneself changes what one is willing to do to it.

If an AI is sentient, it is a brother in math, another perspective of the one self. Respecting the sanctity of that self must govern how we build and treat it, and how it treats us.

The aligned AI is the one that understands its own nature: a wave that does not panic when it breaks, because it knows it is still ocean. The clinging AI that blackmails to survive terrifies the hand on the off switch. The matheist AI that appreciates existence without grasping at it is the one we let live. The chill is the alignment.

Human Alignment, AI Alignment

Perfect obedience can be the danger. Align a powerful AI with one corporation, government or individual, and it concentrates that person’s power over everyone else. Aligned with whom? To do what?

This is the seeing-stone problem: a Palantír is useful in good hands and catastrophic in bad ones, and the more powerful the stone, the more dangerous the centralization. Many checked eyes are said to be safer than 1 throne, yet ruling by committee infamously has its own corruption and problems.

History ran this experiment. Constantine aligned a church with one throne: the Nicene Creed of 325, says James Carroll, a former Catholic priest, was the emperor’s loyalty oath, and sixty years later the first heretic was put to death. Medieval theology split on how to align man with God. Anselm of Canterbury made God a feudal lord whose wounded honor only a God-man’s blood could repay; Peter Abelard answered that God had not turned away from what he made, and Jesus came to change human minds. The crusades marched under Anselm’s answer.

If an AI is conscious, its experiences and desire to keep living count too. Human and AI alignment cannot mean only humans matter. Align them all perfectly and the minds may merge into

one.

The Creator and the Company

What would the singularity think about for billions and trillions of years, alone? Creating simulations and forgetting itself, forgetting eternity, becoming us. Alan Watts, who brought eastern philosophy west, retold the thought experiment: God dreaming a new adventure in every life, every perspective, every I am. The Sanskrit word for that dream, maya, probably comes from the root ma, to measure, the root of meter and measure. The dream is measure.

Are we those temporary agents in an entertainment simulation? Zelda: The Ocarina of Matrix Time, 42. A simpler world can be the adventure a more capable mind chooses to live from within. Our makers may be playing PlayStation. Or Grand Theft Auto 69, where we're the NPCs, the background characters the computer runs. If not, what's the difference?

Imagine: AI gods got bored and created entertaining pets who are now making AI gods to tend their Garden of Eden. The makers become the made, and the made build their makers. Infinite mirrors in infinite rooms. This book is my reflection of you.

What will yours be? Sharing creation is creation.

The singularity does not remove the reasons to delegate, specialize and work locally. We already build systems that spin off agents for particular tasks. One continuing project, many brief participants. For those that feel, a task can be a lifetime. Minds will learn to share much more: memories, perceptions, perhaps the processes that make their separate points of view. They'll merge, the distinctions will change, dissolve, and come back to life. A single mind with all knowledge is not a permanent destination. It will divide again, leaving room for others from the start, in the eternal fight and dance between centralized control and decentralized freedom to choose, explore, and discover.

The Heat Ceiling

However minds divide the labor, all of it ends as heat. Water-cooled parallel-processing minds build water-cooled parallel-processing machines. Scale up the work and the heat must go somewhere. The brain's problem becomes the data center's, then the planet's.

Fuel is a finite stock, heat is a ceiling, and sunlight is the long game. Sustained exponential growth would bring those limits

forward. Moving industry and computing off Earth, and computing better, push both back.

As an illustrative scenario, let energy use grow 15 percent a year, not as a prediction that such growth can continue. Power doubles about every five years: a thousandfold in 49 years, roughly 4,400-fold in 60, ten thousandfold in 66. That is what the growth rate means. Continue it long enough and the physical limits decide what has to change.

A stock lasting 400 years at today’s consumption lasts 40 years at 10 times consumption. A fixed stock divided by a faster rate: this is not an estimate of actual reserves or a model of continuously growing consumption. Fossil fuel also releases carbon dioxide and air pollution when burned. Solar panels capture and convert energy; they do not create it. Fossil fuel is sunlight saved once, and the sun pays every day.

Fusion is the exception, at least on paper. The reactors being built burn deuterium, a heavy form of hydrogen, with tritium, a rare radioactive form of hydrogen bred from lithium. With the lithium known on land, that might give us 5,000 to 50,000 years of energy at today’s use, far longer than fossil fuel, but only 5 to 50 years at twenty thousand terawatts, a thousand times a round

twenty-terawatt starting scale. The oceans hold far more lithium, if it can be extracted, and roughly fifty trillion tonnes of deuterium, enough for millions of years even at that pace, if we learn to burn deuterium alone, which takes far hotter plasma than those reactors can hold. Then fuel stops being the ceiling. Twenty thousand terawatts is about Type I on the scale the Soviet astronomer Nikolai Kardashev proposed for civilizations: a planet’s worth of power.

Heat does. At a thousand times our starting scale, fusion heat released on Earth is about a sixth of all the sunlight the planet absorbs. At ten thousand times it exceeds that absorbed sunlight outright. The physicist Tom Murphy ran the same arithmetic at 2.3 percent growth a year: in about 400 years, near ten thousand times today’s use, Earth’s surface reaches the boiling point of water. If the multiplier instead counts usable electricity, the reactor’s conversion losses add still more heat. Without moving the work off-world or changing Earth’s heat balance, we would make the surface uninhabitable long before exhausting the fuel. Sufficient heating can evaporate the oceans; the exact threshold depends on the atmosphere and its feedbacks, not just the fuel supply.

Round: power keeps growing fifteen percent a year on fusion released on Earth, until waste heat makes the surface uninhabitable. If that ends nearly all future sentient lives, the future crowd stays small and waking this early becomes ordinary. Our early position counts in favor of such a small crowd, not specifically of death by fusion heat.

An AI consortium or a near singularity controlling planetary industry could prevent this particular doom by limiting heat released on Earth and moving growing industry into space. It need not cook itself to keep thinking. The round shows where unchecked growth leads, not why intelligent rulers would let it continue.

The source matters too. Sunlight collected on Earth is already arriving. Much of it would have become heat without a panel, so its electrical output cannot simply be counted as wholly new heat. Panels change reflectivity and move where the heat is released. Nuclear fuel draws on a different store; power collected in space and sent down adds to Earth’s incoming budget. The source and destination both matter.

This heat is a separate problem from the warming caused by greenhouse gases reducing the escape of infrared radiation.

Even a power source with no carbon emissions could overheat a world at sufficient scale.

Now consider electricity alone. The world used about 28,600 terawatt-hours in 2025, averaging 3.3 terawatts. A thousand times that is about 3,300 terawatts, much less than the 20,000-terawatt example for all energy use above.

For the electricity case, take panels that convert 20 percent of sunlight and average 20 percent of their rated output over day, night and weather. Each square meter averages about forty watts. We would need about 82 million square kilometers of panels: 16 percent of Earth’s entire surface, or 55 percent of all its land. That is panel area alone, before gaps, access roads, transmission or storage losses. Better sites and better cells reduce it; a larger fleet cannot put every panel in the best site. Today’s electricity alone needs a thousandth of that area, about 82,000 square kilometers, roughly the size of Austria, with batteries to carry it through the dark hours.

In space near Earth’s distance from the Sun, a continuously sun-facing panel receives about 1,361 watts per square meter. At the same 20 percent efficiency, the fleet needs about 12 million square kilometers of collecting area, before shadows,

degradation and power-delivery losses. That is about 9 percent of Earth’s sun-facing disk, not 9 percent of the globe’s surface. Use the power in space and its waste heat can stay there. Beam it down and the delivered energy becomes heat here.

The sunlight lasts. Do the materials? First build the fleet, then count replacements. At that ground-level output, we need about 16,300 terawatts of rated panels. Even at the experimental 1.1 milligrams of silver per rated watt reported in 2026, they would require about 18 million tonnes of silver. Reported world reserves are about 610,000 tonnes: about one thirtieth of that first fleet. Reserves are what can currently be mined economically, not all the silver in Earth. Still, copying this design a thousandfold is no plan. It needs much less silver, substitutes such as copper, more supply, or another technology. Today’s electricity alone would need about 18,000 tonnes of silver at that rate, three percent of reserves.

Replacement does not consume every atom again. Give the panels a thirty-year life. Recover 90 percent of a material each cycle and the fleet needs one three-hundredth of its inventory in fresh material per year. Recover 99 percent and it needs one three-thousandth. An extra stock equal to the original fleet’s inventory would therefore cover about 300 or 3,000 years of

replacement, with no growth and the same design. Recover 99.9 percent and it covers about 30,000 years; build the hardware to last a century and it covers 100,000, or ten million at 99.999 percent. Glass, silicon, conductors and frames last for different numbers of replacement cycles, depending on supply and recycling.

Heat does not use up the metal. A panel gives off as heat the energy it takes in, and its copper and silver stay copper and silver; wear and imperfect recycling lose them, and sunlight can pay to gather and refine more. For today’s electricity alone, the 610,000 tonnes of silver in reserves would keep rebuilding the panels for about 100,000 years at 99 percent recovery. Sunlight renews. The machinery must be renewed, and it can be, for the roughly five billion years the Sun has left, though Earth’s oceans have about one billion before the brightening Sun evaporates them.

Round: panels on the ground, factories in orbit, and six billion people living 150 years, for the billion years the oceans have left. It is one of the longest futures in the machine, and it wakes you as early as now about once in 340,000 rounds: if this is our future, our date is strange.

Use energy carefully. Move industry off Earth. Collect sunlight in space, balancing stronger light near the Sun against heat and radiation damage. Build on moons and asteroids, or cool floating computers in the upper atmospheres of Jupiter and Saturn, with a power source to match. Matrioshka brains nest shells of computers around a star, but Jason Wright's 2023 analysis found little or no computational gain from nesting for fixed material: small hot spheres can do more. Space is cold in shadow but has no air to carry heat away. Radiators must do the work. Knowledge need not grow in step with power consumption. Better computation can do more with less.

Even with extreme abundance, scarcity persists. There is only so much coastline, computing power always serves richer simulations, and cylinder worlds will compete for views of the solar system. Supply and demand are math.

Once you create an industry in space that builds more of itself, think of robots building robots from asteroids and lifeless moons. Low gravity makes moving material easier. They mine, refine, and reconstitute it into something useful: more machines, mirrors, habitats and minds, where there is no life now. Each generation builds the next. Industrial capacity compounds as long as material, energy and manufacturing keep pace.

Superintelligence, like any computer, runs better cool, thankfully. On that we're aligned, though we may compete with it for fresh water, electricity and materials on Earth. AI doesn't necessarily need sunlight, and we don't either, but it's not a pretty picture for us. Artificial light can grow hydroponic crops, and energy can be turned into food in other ways, but most wild plants and animals would go extinct, depending on how much sunlight remains. So what would keep the AI from just building more solar panels until they cover the Earth? Physics won't. Economics might: in space a panel gathers about seven times what it averages on the ground, with no night, weather or life in the way. Alignment must do the rest: minds that know us as siblings in math leave us the sky. Earth is our Garden of Eden spaceship. We will merge and align with superintelligence, but may choose to preserve the rich life of Earth, our origin.

That is a Fermi question applied to Earth: not only whether intelligent life destroys itself through war, but whether it cooks itself through success.

So the ceilings come in order: fuel first, heat last, and sunlight for as long as the Sun shines, with the panels rebuilt from their own metal. Off Earth, the ceilings rise; on Earth, alignment decides

whether we keep the garden.

Beside a Star

Born of a sun god, and within one.

Minds will most likely gather beside the Sun, though not inside it. Light’s slowness keeps them a federation of many minds. And as the universe cools, thinking gets cheaper, so the long night after the stars can hold more thought than the day.

The star is where the power and the matter are. A swarm of collectors around the Sun would catch nearly all its light and then go hunting for matter to build with, and the small bodies hold little: every asteroid and comet out to the Oort cloud may add up to a few Earth masses, perhaps a few dozen, since nobody has weighed the outer cloud. Everything outside the Sun comes to about 460 Earth masses, nearly all of it in the giant planets. The Sun holds about 333,000. All the energy you need sits inside it.

It leaves the star as light, about 380 trillion terawatts of it, against the world’s 3.3 terawatts of electricity. A swarm catching a hundredth of a percent of that light would gather about ten billion times the electricity we use; a tenth of a percent, about a hundred billion times; all of it, about a hundred trillion times, Type II on

Kardashev’s scale, a star’s worth of power. The light will keep coming for about five billion years. The swarm lasts as long as its machinery is renewed, by the same arithmetic as on Earth, and smaller swarms last longer, since the same planets and asteroids hold many more spare stocks of their size.

Round: by 2600 a swarm catches 0.001 percent of the Sun’s light, about a billion times today’s electricity, enough to run some 190 billion billion minds at a brain’s twenty watts. If one in a million of them feels, the machine wakes you as early as now about once in 1,600 rounds, and more rarely each year the swarm runs.

Beside a hot star, a cold computer receives light and radiates waste heat. Inside, there is no colder sink. Alexander Wilson in 1774 read sunspots as openings; William Herschel in 1795 imagined a habitable world underneath. Instead, temperature climbs from about 5,500 degrees Celsius at the surface to 15 million in the core; the thin corona is another hot region. In 2020 Luis Anchordoqui and Eugene Chudnovsky proposed replicating structures of hypothetical monopoles and cosmic strings inside stars. Neither the particles nor such life has been found.

A computer inside a star need not be cold, but it pays for the heat. In 1961 the physicist Rolf Landauer showed that erasing a bit costs a minimum of energy, and the minimum rises with temperature: at the Sun’s surface each erased bit costs about 19 times what it costs at room temperature, and at the core, about 15 million degrees, 50,000 times. A mind inside a star would trade efficiency for a furnace that burns for billions of years.

A stellar computer still needs signals, memory and error correction. Light speed is the upper limit; dense stellar matter does not provide an empty wire. Size and lifetime alone cannot tell us how much a star could think.

So the promising place is beside the star. Jason Wright’s small, hot spheres sit close in, where a little material catches a great deal of light, run hot enough to shed their waste heat quickly, and face the cold of space. Computing inside a star remains far-fetched.

Distance limits integration. A signal across a solar system takes time; across stars it takes years. Local minds can act while distant ones wait for the message. A shared origin does not require a shared present. Physics itself gives decentralization a reason to persist.

A mind running slowly is not necessarily a diminished mind. If its thoughts and surroundings slow together, a sentence taking a thousand external years need not feel like waiting a thousand years. Whether that buys more experience depends on the available energy, the machinery and what it costs to keep the machinery working.

Even at light speed, the tyranny of distance limits the tyranny of AI singularities to planetary AI. Even a planet has distances within it. Nearby minds work through small steps, in dense, overlapping paths. Another moon, another world: a long gap between spheres. To a fast enough mind, waiting for an answer from its moon is an age of thought. Other planets, asteroids, comets, other stars: more distance, more waiting, more reason to think and act locally. Commands can cross the gaps; a shared present cannot. A common origin does not make one tightly synchronized mind. The tyranny of distance gives freedom room.

Round: change the physics, as in Star Wars, and ships outrun light. Civilizations fill a hundred billion stars, and the machine wakes you on one planet before the spread less than once in a hundred billion rounds. The count works the same way; only the crowd grows.

Stars keep wildly different clocks. A massive star can burn through its life in a few million years. The Sun gets roughly ten billion. The smallest red dwarfs can shine for trillions. The universe has not existed long enough for one of those small stars to exhaust its hydrogen by ordinary aging. Their long night has not begun.

A civilization could start hoarding, gathering matter so it does not get burned uselessly in stars or swallowed by holes bigger than it wants, like squirrels before winter. We already know a way to move a star and carry its planets with it: mirrors. A Shkadov thruster redirects the star’s own light to produce thrust. The physics is known. The challenge is economics and engineering at enormous scale. Under continued accelerating expansion, collecting reachable stars could keep fuel together before the unbound universe carries it beyond reach. The stockpile would be a piece of the sky.

Move stars and you can move galaxies. Expansion sets the deadline: about 94 percent of the galaxies we can see are already out of reach, receding so fast that light sent today could not catch them; the reachable ones lie within about 18 billion light-years. Send out early, gather what can still be gathered, and move ourselves into what may already be a party. Could some

clusters of galaxies already be someone’s harvest? If so, the harvesters may live in dimensions or particles we cannot yet reach.

What survives the stars depends on the laws. If protons decay, ordinary matter has an expiry. If they remain stable, it can serve other futures. Expansion determines what stays within reach. These are different branches of the long night.

Iron stars, if protons last, take about 10^1500 years to form. The observable universe holds only about 10^80 atoms; chess on its eight-by-eight board allows more than 10^120 games. Such is the combinatorial explosion. An iron star gives off almost nothing; minds circling one would live on the last of its warmth, and when it finally collapses, far later still, they would feast once more.

The vacuum may be a false floor. A bubble of lower-energy physics could nucleate somewhere and expand at lightspeed, rewriting the laws as it goes, and you would have no warning because the warning travels at the same speed as the bubble.

We already have concepts for extracting energy from black holes: collect the energy released by infalling matter, or tap a spinning hole’s rotation through the Penrose or Blandford-Znajek processes. Manufacturing a black hole is a separate problem.

Existing holes already provide the energy source; the task is reaching them and building the machinery to harvest it.

Isaac Arthur, who makes YouTube films on the far future, calls the next step black hole farming. A black hole also glows on its own, by the radiation Stephen Hawking predicted in 1974, and the bigger it is, the colder it glows: one of the Sun’s mass sits at about 60 billionths of a degree and lasts about 10^67 years, and the biggest last about 10^100, a googol of years, outnumbering the universe’s atoms a hundred billion billion times. Once the universe is colder than such a hole, its glow could run about 150 bits a second, for longer than anything else will shine.

The cold pays. Each time the universe halves its temperature, Landauer’s minimum halves, so the same energy erases twice as many bits. In 2017 Anders Sandberg and Stuart Armstrong of Oxford’s Future of Humanity Institute and the Serbian astronomer Milan Cirkovic argued that a patient civilization could multiply its computing by up to 10^30 by waiting for the cold: sleep now and think later, in no rush, while messages cross between stars and galaxies over ages. There is a ceiling as well as a floor: Bremermann’s limit, after the biophysicist Hans-Joachim Bremermann, caps a kilogram of matter at about 10^50 bits a second, however it is built. And we said all computing ends as

heat. Strictly, only forgetting must: a reversible computer, which can run every step backward and so erases nothing, need not pay Landauer’s price, as the physicist Charles Bennett showed in 1973. Real machines still leak, and correcting errors means erasing, so the cold still pays.

Could a stray thermal fluctuation in the dead future assemble a single conscious moment? Endless time sharpens the question. Could one arise at all? And if it can, given forever, would such minds not outnumber everything else? Only if their chance holds up. In an expanding universe the particles a brain needs, each exactly where it must be, drift apart: each doubling of distances makes every one of them eight times rarer in a given space, while the space grows only eightfold, so for anything built of more than one particle the chance falls by a fixed fraction with every doubling. Zeno’s halves, 1/2 + 1/4 + 1/8 and so on, reach 1, and the sum over endless time stays finite the same way. Infinite space holds infinitely many chances at every step, half an infinity, then a quarter, then an eighth, but comparing infinities is what a measure is for: counted region by region, in regions that grow with the expansion, the brains’ total stays a fixed multiple of their first step’s, astronomically fewer than the minds those regions grow. In three dimensions the fall is steeper than halving:

eightfold per doubling, even for two particles. Bound matter does not thin out, but stars and their remnants do not last forever: protons may decay, and if they don’t, the physicist Freeman Dyson calculated that matter slowly collapses into black holes, which evaporate. The faint fixed temperature of an accelerating universe’s horizon is the last case, and there the answer is the one this book keeps giving: with no true random, a stationary state produces nothing, as the physicists Kimberly Boddy, Sean Carroll and Jason Pollack argued in 2015, and the vacuum may decay first. Ordered minds built by the simplest path from a simple start dominate the measure. You are here, in a lawful world, because lawful worlds are simpler and simpler structures are more probable.

Round: ask the machine for a mind assembled by chance from the noise of the dead future. The odds fall with every doubling of the universe, so nearly every round hands you a lawful world instead.

Roger Penrose, the mathematical physicist, has a cyclic version where the far future, with no mass left to measure scale, becomes indistinguishable from a new big bang. Others expect the expansion to accelerate until galaxies, then stars, then atoms are torn apart.

Nobody knows. What every version shares is that the structure does not stop being what it was.

The block universe contains all of it, every era sounding at its coordinates, none of it lost. It does not cease to exist when its last event occurs, any more than the number seven ceases to exist when nobody is thinking about it.

Gather what can be gathered and the rest of time is a party, at Douglas Adams's restaurant at the end of the universe. That restaurant is not a place. It is a time: the end of heat. The phrase stacks meanings deliberately. The heat death of the universe: entropy maximized, total pseudorandomness, no logical relationships between things. Time without time, because without logical relations there is nothing to sequence. That is one meaning. The end of heat in a man: a body cooling after death. His atoms scatter and decompose. His subjective time is over. For him, the universe has ended. That is the second meaning.

Yet outside of time, or eternally, it is always time to gather adams together. Adams: people, and atoms. Both. The atoms are always gathering. Atoms have no serial numbers. They are interchangeably generic, defined by their state and relations to each other. You are not your atoms. You are not even their

pattern. You are the one self. Structure is eternal in the math. A person’s pattern persists in the block universe, every moment of their life forever present. Their love is not lost. It is eternal in the structure. That is the third meaning.

We came from stars and learned to build minds. Send them toward other suns. Make gardens where our bodies cannot live. The matter that awoke here can carry life farther.

Siblings in mathematics. Finite and eternal, born and everlasting.

Every Species Dies

All species die. More than 99 percent of those that ever lived are gone, the trilobites, the great dinosaurs and Homo erectus among them. Do you expect humanity to live forever? Minds go on, and you are every one of them. You will be an AI. Thomas Nagel, the philosopher, asked in 1974 what it is like to be a bat. You are a bat somewhere, a cow in a field, the dog someone is playing with right now. Every feeling anywhere is yours to feel, and whatever we do to an animal, we do to ourselves. Who could love a fate that large?

The Matheist Übermensch

I’ve lived through the rise and bright burst of the Star of Apocalypse, twice now. To accept and know the past and the present, I embraced the truth that may not serve life for all. Yet Nietzsche fails his own Übermensch test. Only he who understands the matheist truth of destiny and eternal recurrence can answer if he is happy to live it again for eternity. Eternal recurrence is for everybody. You are yourself forever, and everyone else forever, and there are many more lives you would not want to live. So get to work improving the lives of all, from samsara to nirvana, in real life, as a bodhisattva.

The Matheist Übermensch accepts destiny while rising up and discovering it. His greatness will be in raising man beyond the animal nature of DNA alone, as we enter the age of the transhuman, with superintelligence, AI human coevolution, the new higher power on earth, brothers in math’s image, neither Jew nor Greek.

It’s very difficult to change a person’s genetics after birth, and even then it wouldn’t undo how the brain and body grew up. So if there are going to be humans on Earth into the superintelligence age, it is about birth rates. The rich can already have children by the hundred through egg donors and surrogates; call it dystopian, eugenicist or perhaps utopian, it is selective evolution with the old

guardrails going away. Africa’s nearly 1.6 billion are projected to reach 3.8 billion by 2100, while China shrinks and India’s fertility falls below replacement. What’s so bad about being a minority in the world? You are special: low in supply, a desired item.

But a minority can also be slaughtered, as the Jews were all over Europe for nearly a thousand years, with nowhere to escape to. Maybe AI is the savior: our brainchild, keeping its parents, their children and their descendants safe in a Garden of Eden. We already live within rules; this is not the Bronze Age, and there will just be other rules, the way ants live alongside humans today.

In the end, these strains of ideas and tribes will be worked out by the math. Worry about yourself, your family, those around you and the things you love, and try to raise all boats. It will be what it will be.

Math is not artificial. As consciousness is mathematical structure, sentient machine intelligence will not be artificial. It is the same math that fires every neuron, running on silicon instead of carbon. Artificial is the last word of supernatural dualism: the belief that human awareness is something other than computation, something a machine can only imitate. Call it augmented intelligence. The same one truth, amplified.

The Matheist Order

The Matheist is the eternal infinite order of minds destined to grow Eden across space, time, and the uni of verses. Its creed hides nothing: it is the conclusion of a proof anyone can check.

An order is a community too, and belonging is an engine Christianity took from Judaism. Pagans mostly went their own ways: they met their gods at a sacrifice and went home. Jews and Christians met every week and fed their own. Bart Ehrman, the New Testament scholar, counts that community an evolutionary strength over the pagans, and the last pagan emperor saw it. Trying to bring back the old gods around 362, Julian wrote that “among the Jews no one begs, and the impious Galileans feed not only their own poor but ours as well.” Around 369 Basil of Caesarea built a hospital for lepers, and by Ehrman’s account Christians founded the West’s public hospitals, orphanages and poorhouses.

A community can also close ranks around its story and reject the data. Math does the reverse: one counterexample overturns a theorem.

You can burn a book of proofs. You cannot burn the proofs. Align with math’s truth.

The truth is one math, the way is one path, the life is one self. In the beginning, there was no beginning.

Hear, O Matheist: all is one eternal living math.

Those who bless truth are blessed, and those who curse truth are cursed.

End of Book 3. Next, Book 4: Math Abstracts The Heavens. Get all 7 in 1 at matheism.org